Smarter Device Financing Decisions

A better risk model and a simple in-store tool for Verizon

Proposal to the Verizon Steering Committee

November 2026 | The Outliers | Costantinos, Obaro, Stephen

Executive summary: Verizon is turning away customers it would profit from; our model and tool can finance them safely

+$21M

More profit per 1 million applications (+17%)

Today's denied and counteroffer applicants default at about 17%, far below Verizon's 41.7% break-even.

0.620

A sharper risk model that stays explainable

Ranks risk better than the current score (AUC 0.620 vs 0.590) and predicts today's default rate correctly.

3

A simple in-store answer: approve, test or deny

Four facts reps already collect give one consistent decision in seconds. No more counteroffers.

Our ask: approve a 16-week pilot that starts in shadow mode, with no change to customers.

Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis

What we heard: since May 2024, more customers default and Verizon has tightened, but the decision process has not changed

The situation

- **300,000 applications** in 2023 and 2024, decided in store
- Reps sort each applicant **by the internal risk score:** approve, counteroffer or deny
- May 2024: defaults rose **6.8%** → **10.7%** for the same kind of customer
- Verizon tightened: **approvals 89%** → **71%**
- **18% of applicants** now leave without a plan

What Verizon needs from this proposal

1. **A better risk model**
   More accurate than the current score, and right for today's conditions
2. **A live, low-risk decision tool**
   Fast and consistent in every store, and safe to roll out
3. **The best return on investment**
   A credible profit estimate for 1 million applications

The question: how many more customers can Verizon safely finance, and how should stores decide?

Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis

Each plan earns 25% of the device price if repaid and loses 35% if not, so a plan only loses money above 41.7% default risk

Four outcomes of every decision ($798 phone)

	Customer pays	Customer defaults
Offer plan	**+$200** Profit: 25% of price	**−$279** Loss: 35% of price
Deny plan	**−$200** Missed profit (opportunity cost)	**$0** Loss avoided

- **Offer whenever p < 0.25 ÷ 0.60 = 41.7%**
- **Counteroffers** lose 35–50% of customers and don't lower default

Expected profit per plan, $798 phone, today's conditions



Judge approvals by profit, not the default rate: approving more raises both defaults and profit.

Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis

We chose an explainable scorecard: as accurate as machine learning, and simple enough to run and explain in a store

Current internal score — AUC 0.590

- Doesn't adjust to today's higher defaults
- Rewards existing customers, who default more

Gradient boosting — AUC 0.621

- Hard to explain to customers and regulators
- No more accurate than our scorecard

Our scorecard (chosen) — AUC 0.620

- Each factor has a clear, auditable effect
- Gives a default probability to compare with 41.7%

What goes in, and what the data showed

Credit score
Lower score, higher risk; little extra effect above 650

Income
U-shaped: risk is higher at both the lowest and highest incomes

Age
Two steps: under 25 (1.43× odds) and 67+ (1.48× odds)

Existing customer
1.22× odds of default, not a bonus

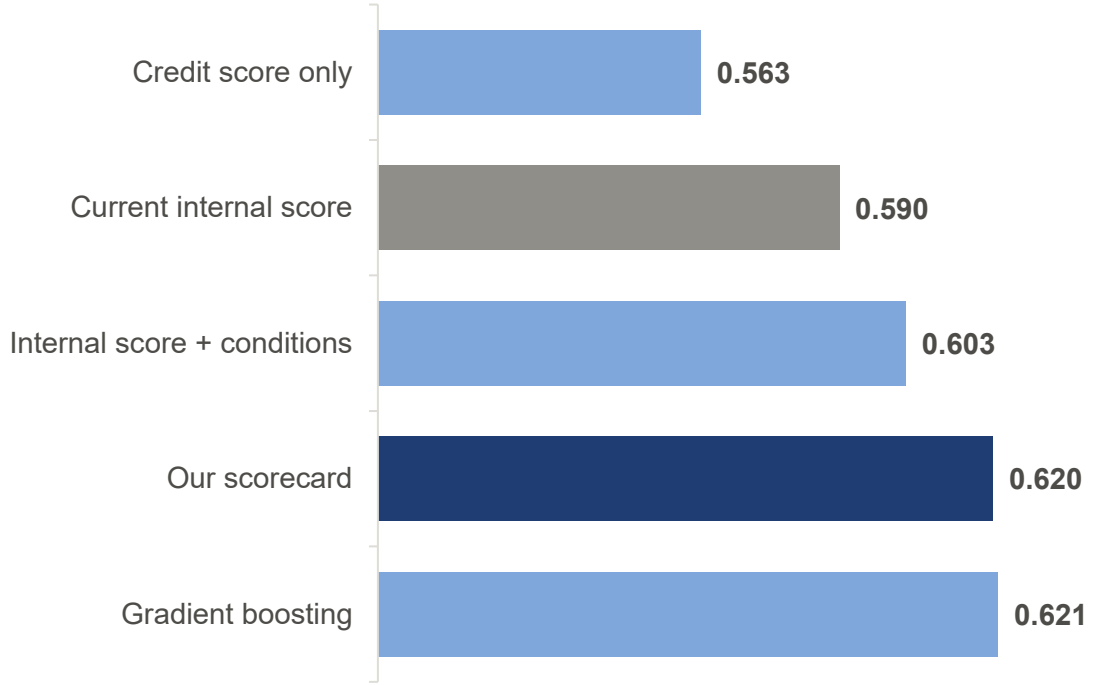
Current conditions
1.65× odds since May 2024; updated each quarter

Verizon gets machine-learning accuracy from a model every rep, manager and regulator can follow.

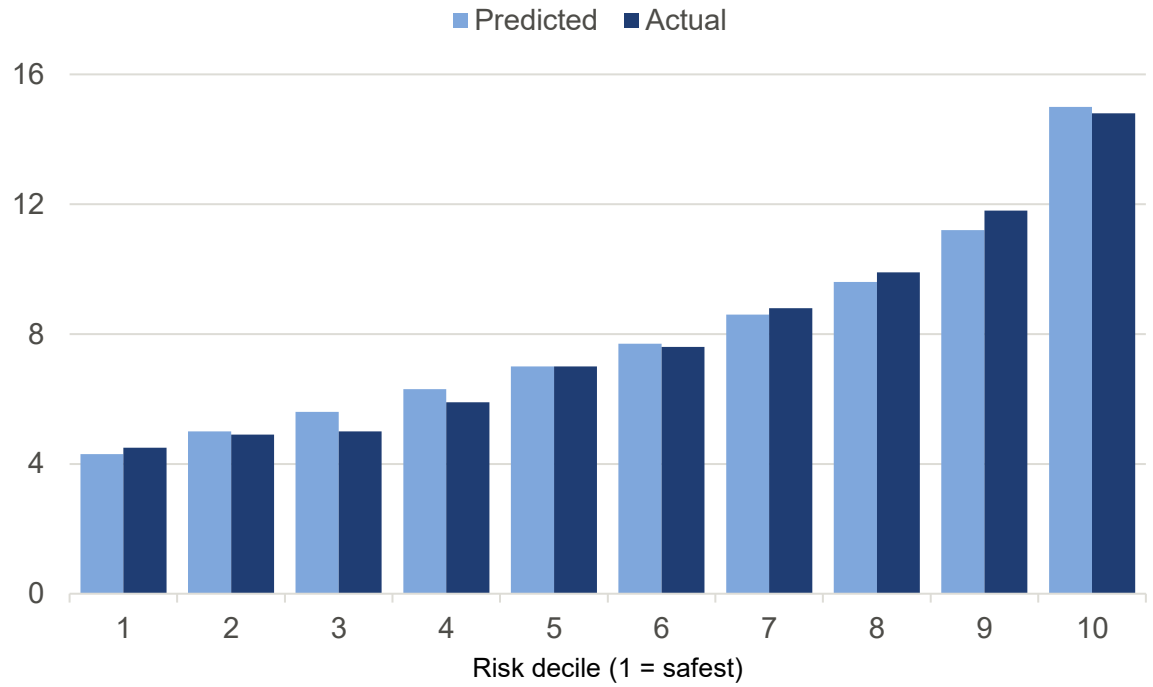
Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis

On 79,518 held-out plans, the scorecard ranks risk best and predicts today's default level correctly

Ranking power (AUC, higher is better)



Predicted vs actual default rate (%), by risk decile



For today's conditions it predicts 10.7% default, matching actual; the current score predicts 8.6%.

Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis

One risk estimate replaces three score bands and the counteroffer step

Today (since May 2024)

71% — **Approve**
Requested phone and terms

17% — **Counteroffer**
Cheaper phone or bigger down payment; 40% walk away

11% — **Deny**
No outcome data, so no way to learn

Proposed: decide on predicted default risk (p)

~97.6% — **Approve on requested terms**
p up to 25%: the range already funded today

~2.4% — **Test lane**
p 25% to 41.7%: approve a random 10% to learn

<0.1% — **Deny**
p above the 41.7% break-even

Financed applicants rise from 82% to about 98%, and no one is pushed into a phone they didn't ask for.

Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis

In store, the rep enters four facts and gets one clear answer with the reasons behind it

Device financing check

Credit score	**575**
Annual income	**$42,000**
Age	**23**
Existing customer	**Yes**

APPROVE on the requested phone

Estimated default risk **18%** (approve up to 25%)

Main factors: under 25, credit score 575, existing customer

Why this is a low-risk system to run

1. **Uses data stores already collect**
   Credit score, income, age and customer status; no new checks
2. **Consistent and fast**
   One formula, the same answer in every store, in under a second
3. **Explainable**
   Every decision shows its main factors, for the rep, the customer and audits
4. **Controlled centrally**
   Cut-offs and the conditions term change at head office, with no store retraining

Same answer in every store, in seconds, with the reasons logged for every decision.

Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis

Per 1 million applications, the new process adds about $21M of profit (+17%)

$$\Delta Value = \Delta Profit - \Delta Loss$$

$$= \Delta(Paying\ Customers) \cdot (Customer\ LTV) - \Delta(Defaulting\ Customers) \cdot (Cost\ of\ Default)$$

Per plan: $(1 - p) \cdot 0.25 \cdot Price - p \cdot 0.35 \cdot Price = Price \cdot (0.25 - 0.60 \cdot p)$

chance of paying × LTV of 25% of Price — chance of default × cost of 35% of Price — break-even at p = 41.7%

p = chance the customer defaults, from our scorecard under today's conditions

$Price$ = price of the phone the customer asked for

Per 1M applications vs today	Δ Paying	ΔProfit	Δ Defaulting	ΔLoss	ΔValue	Evidence
A. Fund counteroffer walk-aways on the phone they asked for (68k)	+58.0k	+$11.8M	+10.0k	$2.8M	**+$8.9M**	Data now
B. Give accepted counteroffers the phone they asked for (105k)	same	+$2.3M	same	$0.6M	**+$1.7M**	Data now
C. Approve denied applicants up to 25% risk (90k)	+73.6k	+$14.7M	+16.6k	$4.6M	**+$10.1M**	Pilot confirms
Total	**+131.6k**	**+$28.8M**	**+26.6k**	**$8.1M**	**+$20.8M**	

Profit per 1M applications

$119.7M → $140.4M

+17%, about $21 per application

Default rate of financed plans

10.7% → 11.7%

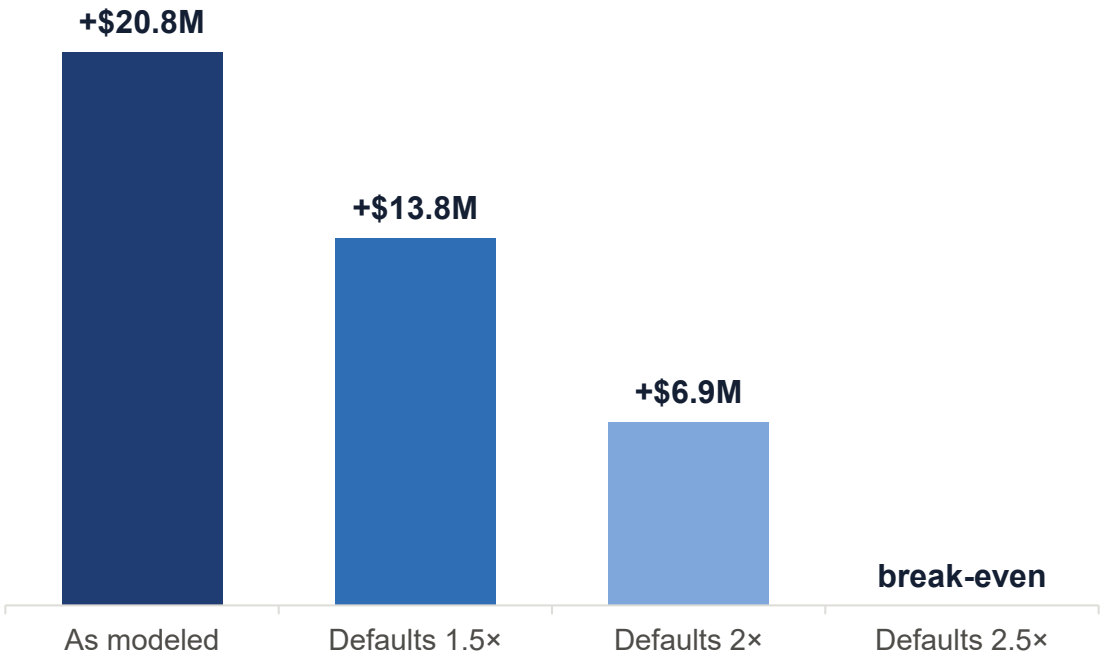
Not counted: service revenue on new lines

Losses rise $8.1M, but profit rises $28.8M. $10.7M is supported by the data today; the pilot confirms the rest.

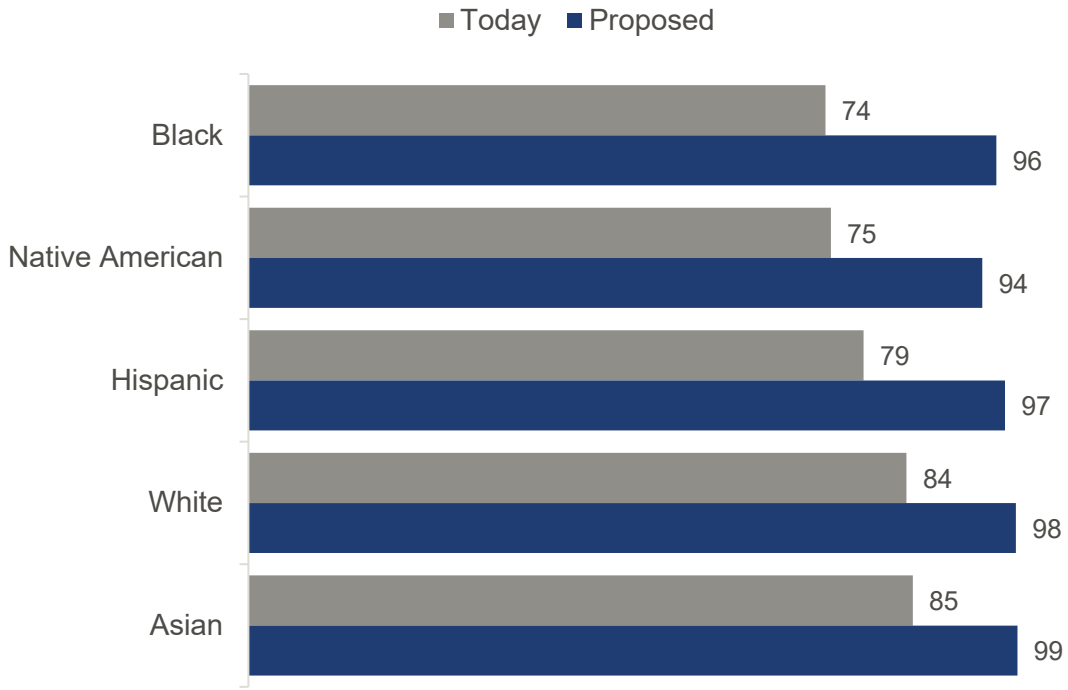
Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis

The gain holds even if defaults double, and it narrows approval gaps between groups

Added profit per 1M applications, if defaults are higher



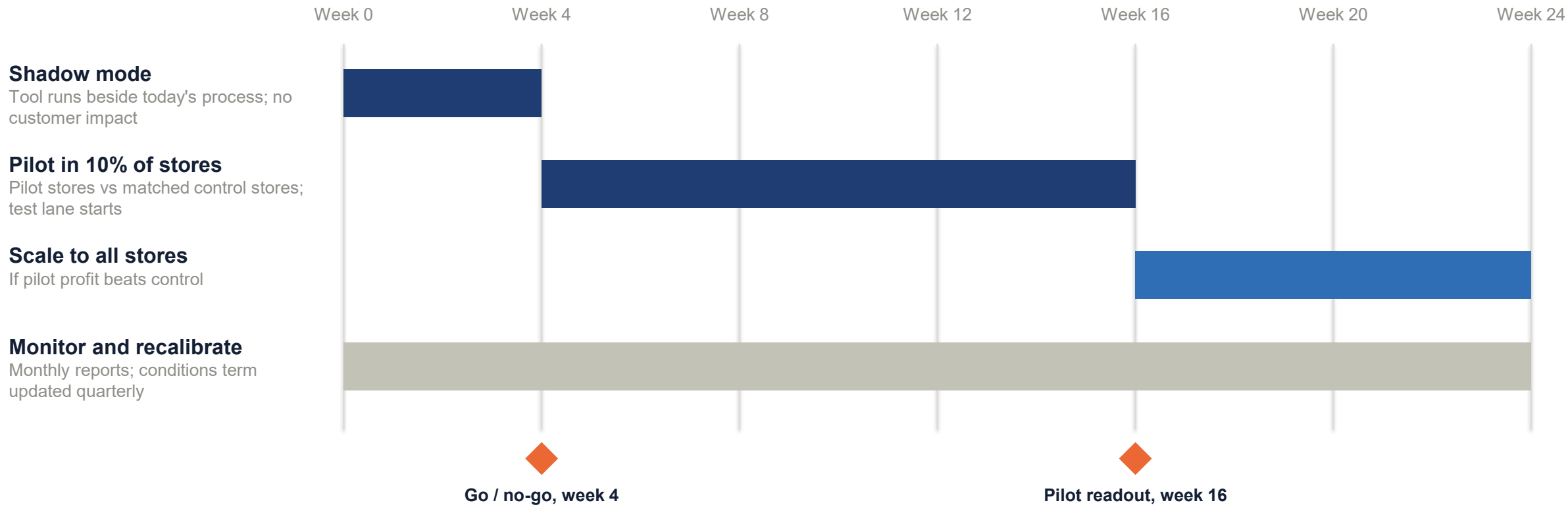
Share of applicants financed (%), by self-reported group



Profit only turns negative if new customers default 2.5× more than the model expects.

Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis

A 16-week rollout tests the tool safely before it reaches every store



Guardrail: pause new approvals automatically if they default above 30% (expected: about 17%).

Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis

Why our proposal: clear value, a safe path to get it, and decisions Verizon can explain

+$21M

Clear value

- Per 1 million applications (+17%)
- Built on Verizon's own profit rule
- Still positive if defaults double

16 wks

Safe path

- Shadow mode first, then a controlled pilot
- Automatic stop-loss guardrail
- Test lane learns about never-funded risk

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Explainable and fair

- Four inputs, no demographic data
- Reasons shown for every decision
- Narrows approval gaps between groups

What we need from Verizon

An owner in Credit Risk and in Retail Operations · IT support to add one formula to the store sales app · monthly repayment data for monitoring

Today's ask: approve the 16-week pilot, starting with 4 weeks in shadow mode.

Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis

Appendix

A. Model details B. Value calculation C. Fairness and definitions

Appendix A: the scorecard is a logistic model with five inputs, fitted on 265,060 funded plans

Term	Log-odds	Odds ratio	Why it is in the model
Credit score (spline, 5 df)	curve		Risk falls with score, then flattens above about 650
Log income (spline, 5 df)	U-shape		Risk falls to about $85k, flat to $130k, rises at the top
Age under 25	+0.356	1.43	Default 11.9% at 18–24 vs 7.5% at 25–66; sharp step at 25
Age 67 and over	+0.391	1.48	Default 9.4% at 67+; step at 67
Existing customer	+0.202	1.22	Existing customers default more (9.1% vs 7.2%)
Current conditions (since May 2024)	+0.502	1.65	Repayment shock that hit every risk band
Promotion period (Aug 2023 to Apr 2024)	−0.083	0.92	Used only to fit the model; not used for today
Tested and left out			Base down payment (a formula of credit score), device tier, internal score (+0.001 AUC)

Holdout: 70/30 split stratified by period and default. Bootstrap 95% CI of AUC gain vs the internal score: +0.024 to +0.038. Within the current period alone: 0.578 vs 0.572.

Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis

Appendix B: how the $21M per 1 million applications is calculated

Item	Since May 2024	Per 1M applications	If defaults 2×
Applications	100,280	1,000,000	
Profit today (actual)	$12.0M	$119.7M	
Counteroffer group on requested terms (6,815 new + upgrades)	+$1.07M	+$10.7M	+$4.8M
Denied applicants up to 25% risk (9,048)	+$1.01M	+$10.1M	+$2.1M
Total added profit	+$2.08M	+$20.8M (+17%)	+$6.9M
Share of applicants financed	81.9% → 97.7%		
Default rate of financed plans	10.7% → 11.7%		
Average risk of newly approved customers	16.8%		

Assumes price of the requested phone, Verizon's +25% / −35% profit rule, and that approved applicants accept (all approvals were funded in the data). Excludes the test lane and tool costs: every $1M of build and run cost is repaid by about 48,000 applications.

Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis

Appendix C: fairness check and definitions

Group (self-reported)	Financed today	Proposed	Model calibrated?
Black	73.9%	95.8%	8.5% vs 8.8%
Hispanic	78.8%	96.9%	8.2% vs 7.8%
White	84.3%	98.3%	7.9% vs 7.8%
Female	81.9%	97.8%	8.0% vs 7.9%
Male	81.9%	97.7%	8.0% vs 8.1%

Calibration: predicted vs actual default rate on held-out plans. Race, ethnicity and gender are not model inputs; they are used only for this check.

Term	Definition
Funded	Funded_Original + Funded_Revised (265,060 plans)
Not funded	Denied, or declined the counteroffer (34,940)
Periods	Baseline Jan–Jul 2023; promotion Aug 2023–Apr 2024; current May–Dec 2024
Profit rule	+25% of price if repaid, −35% if defaulted (every row)
AUC	Chance a defaulter is ranked riskier than a non-defaulter

Source: Verizon device financing applications, Jan 2023 to Dec 2024 (300,000); team analysis